

— THE NATIONAL CENTER FOR AUTOMOTIVE SCIENCE & TECHNOLOGY

The cars that never passed.

Cross-referencing twenty million emissions inspections against the state's vehicle registration record — to find the vehicles that failed, and got on the road anyway.

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— WHO BUILT THIS

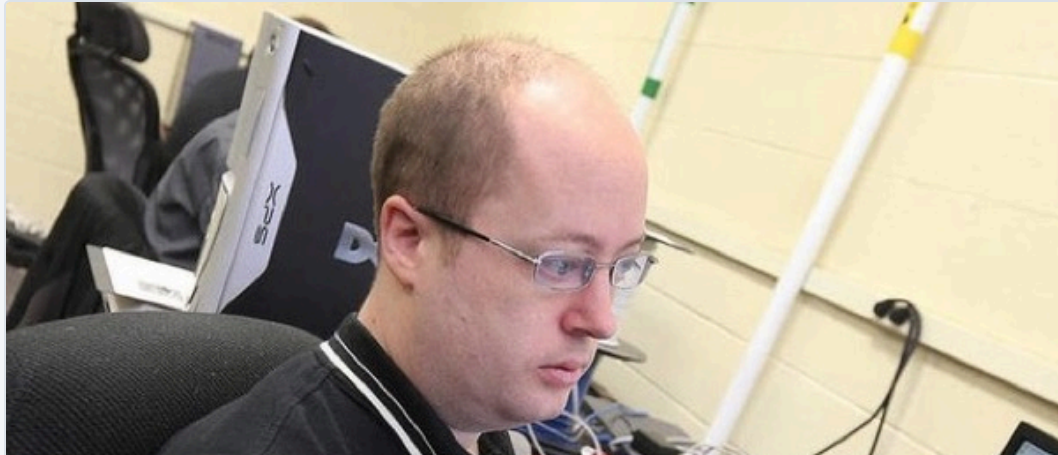
The team



NCAST



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A failed test isn't the end of the story.

Utah requires an emissions inspection in five counties. The counties next door don't. When a vehicle fails, three things can happen next — and until now, nobody could see any of them.

01 It gets registered anyway.

Failed in Weber. No later passing test on record. Re-registered in a county with no I/M requirement — legally on the road, still dirty.

02 It "passes" somewhere else.

Fails at one station, then passes at a station in a different county days later — with no repair in between. County-hopping for a certificate.

03 It simply disappears.

A test that aborts or rejects and never resolves. No pass, no fail, no follow-up. The vehicle falls out of the record entirely.

WHY IT STAYED INVISIBLE

Inspection data and registration data live in separate systems, with no shared key anyone was querying. A county can see the tests it ran. It cannot see what the vehicle did next — or what it did the week before, one county over.

4 days later

THE SOLUTION

Join the records. Then explain the verdict.

One database keyed on VIN, holding both feeds. A transparent rule engine scores every flagged vehicle — and every point it earns maps to a sentence a person can read.

~20M

Inspection records

5

I/M counties

9

Scored risk factors

NOT A MODEL — A RUBRIC

An analyst can always answer *"why is this one flagged?"* with a concrete sentence naming the county, the dates and the station — never "the model says so." Every factor below is auditable in one file, and tunable.



THE SCORING RUBRIC

What you're about to see.

Four views, scoped to a single county, over a rolling twelve months.

- **Fraud-Risk Score**

The ranked worklist. Every score expands into the reasons that produced it.

- **Failed, then Registered**

Vehicles registered after a failed test with no later pass — and the county it was registered in.

- **Evasion Cases & Stations**

Cross-county fail-then-pass cases, and the stations that keep turning up in them.

- **Unknown Outcomes**

Tests that never resolved to a pass or a fail.

→ **Let's look at Weber County.**